

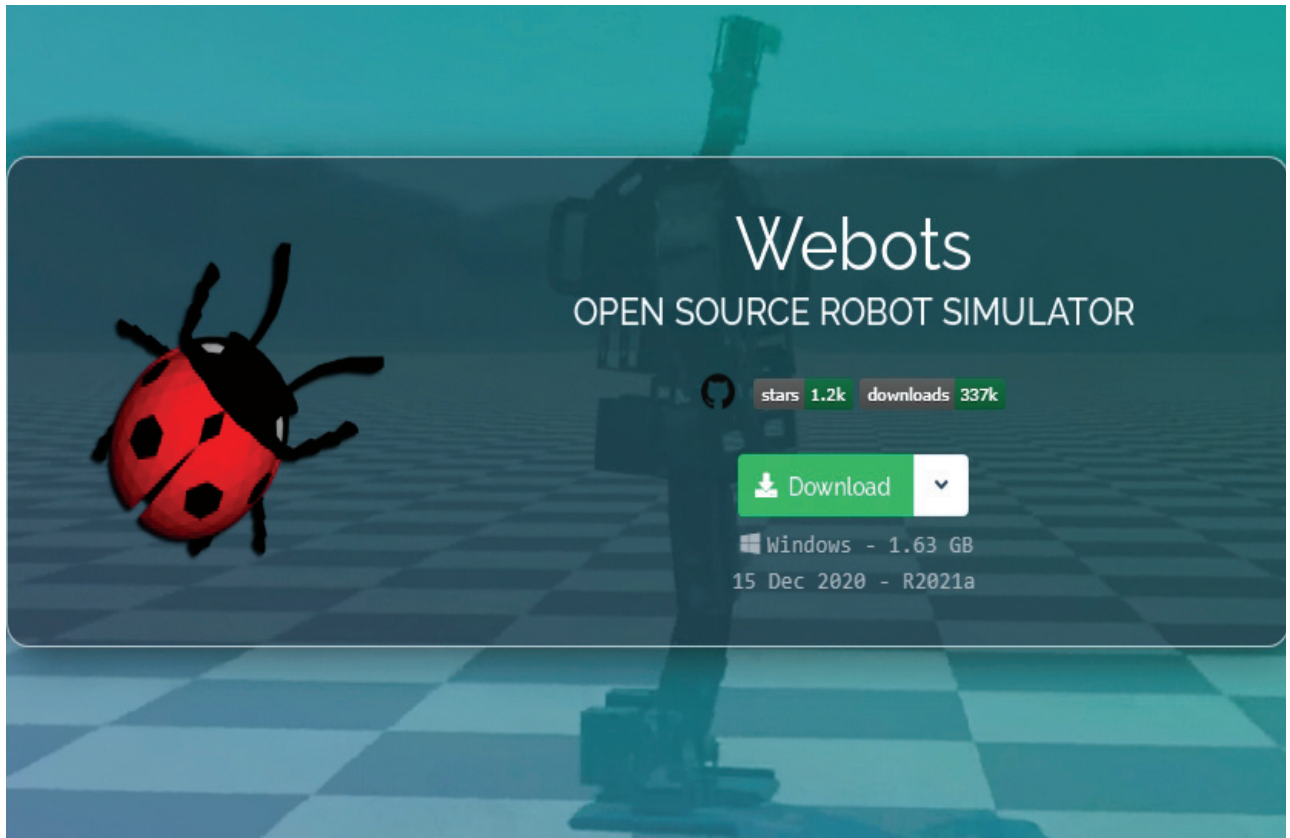


Figure 2: Collision detection in the CoppeliaSim robotic platform



Figure 3: Setting a panel to detect the distance for avoiding a collision

This tool converges dynamics and physics, and uses bullet mechanics, differential equations and other integrations for easy and effective calculations that simulate real life scenarios including collision reaction, grabbing, collision response, and grasping. Multiple recordable data sources (including user-dependent) can view or combine time graphs with xy-graphs. 3D curves can

also be achieved effectively. CoppeliaSim integrates features for real-time automations including proximity sensors, vision sensors, and viewable objects.

Collision detection and avoidance in CoppeliaSim

If a robot is developed for industrial and corporate applications, it should be programmed in such a way that it does not collide with other components or machinery in the industrial unit.

The collision detection and avoidance module is easy to program in CoppeliaSim. The distance and collision settings can be set up effectively in *Menu->Tools->Calculation*.

Researchers and scientists can carry out research tasks using open source platforms and tools like CoppeliaSim for robotic applications, whenever the real world infrastructure and devices are too costly to work with. **END** 🐧

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